TRIAL, ERROR AND TRIUMPH

'See, it fired.'

Some of the early errors at TERLS spoke of our childishness. Once while working on the Judi-Dart project, Kalam was discussing with me how to test the dart. Rocket Judi came from France and I was in charge of developing the dart, which is the pointed portion on the top that holds a bundle of needle-like copper chaff that has to be dispersed in the upper atmosphere through a controlled explosion by opening of the bottom part of the dart at high altitudes. Radars would then pick up the movement of the copper chaff which would float and swim in the upper atmosphere, to understand wind patterns and more.

We had no facilities for simulations or ground-testing. And TERLS director Murthy had given strict instructions not to fool around with explosives on the church campus in Thumba. But somehow we had to do it. When neither Kalam nor I could come up with a method, our draftsman, Alex, casually suggested that we keep the sharp edge of the dart in the beach sand in our backyard and fire it, so that the base facing upward would give way and throw the copper chaff into the air.

Primitive as it may sound now, but we decided to do that. Alex felt proud that the scientists approved his brilliant idea, and off we went with the dart to the beach. We buried the sharp end in the sand and fired. As expected, the copper chaff was ejected into the air. What we didn't know was that the chaff would fly in a trajectory and fall across the road on which Murthy was passing by in his car. I haven't asked him till now whether he felt something falling on his car. When we came back to see the success of the experiment, the dart was missing. In the recoil, it had penetrated the sand. Kalam and I frantically tried to dig and search, but it was nowhere to be found. So, another secret of the Indian space programme lies buried somewhere deep in the sands off the Arabian Sea—the sharp end of a Judi rocket dart.